FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 06-06-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total **54** questions.
- Each subject (PCM) has **18** questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section – B & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases

Section-B (07 – 12, 25 – 30, 43– 48) *contains* ***18 Numerical*** *based questions with* ***Single digit integer*** *as answer, ranging from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-C (13 – 18, 31 – 36, 49– 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

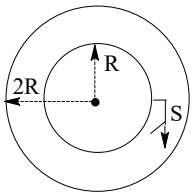
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two thin conducting shells of radii 'R' and '2R' are shown in the figure. The outer shell caries as uniform charge +Q and inner shell is neutral. The switch is 'S' connecting the inner shell to earth then:



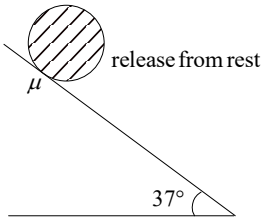
(A) When switch is open potential of inner sphere is same as that of at a point having distance $\frac{3R}{2}$ from centre

(B) By closing the switch S capacitance between the two spheres increases

(C) When switch is closed potential of inner sphere would be zero

(D) When switch is closed charge on inner sphere is $-\frac{Q}{2}$

2. A uniform body with circular cross-section released from rest on rough and fixed inclined plane shown in figure, then which of following is **INCORRECT** for the motion of object.



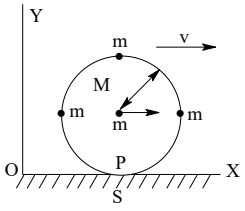
(A) If the body is hollow sphere and $\mu = 0.4$ then it will perform pure-rolling.

(B) If the body is hollow sphere and $\mu = 0.5$ then ratio of rotational kinetic energy to the translational kinetic energy at bottom is $2/5$ forgiven sphere.

(C) If the body is solid sphere and $\mu = 0.1$ then the body will perform pure-rolling

(D) If the body is solid sphere and $\mu = 0.5$ then ratio of rotational kinetic energy to the translational kinetic energy at bottom is $2/3$ forgiven sphere.

3. A uniform circular disc of mass M, and radius R has four particles (each of mass m) rigidly attached to it. The disc rolls on the surface S without slipping. At the instant shown the centre of the disc has velocity $v_0\hat{i}$.



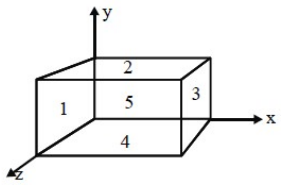
(A) The kinetic energy of the disc particles assembly is $\left(\frac{9m}{2}+\frac{3M}{4}\right)v_0^2$

(B) The velocity of the centre of mass of the assembly is $\left(\frac{M+5m}{M+4m}\right)v_0\hat{i}$

(C) The angular momentum of the assembly about the point of contact P is $3/2(M+6m)$ $\left(Rv_0\left(-\hat{k}\right)\right)$

(D) The maximum speed of a point on the disc with respect to S is $2v_0$

4. In a region of space, the electric field $\vec{E} = E_0 x\hat{i} + E_0 y\hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



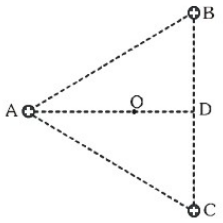
(A) The total electric flux through the faces 1 and 3 is $E_0 a^3$

(B) The charge inside the cubical volume is $2\varepsilon_0 E_0 a^3$

(C) The total electric flux through the face 2 and 4 is $2E_0 a^3$

(D) The charge inside the cubical volume is $\varepsilon_0 E_0 a^3$

5. Three identical point charges are placed on the vertices of an equilateral triangle ABC as shown in the figure. Consider a median AD, where point O is the centroid. As we move from point A to point D



(A) Electric field vanishes only once at O, between points A and D.
(B) Electric field vanishes two times, first at O and then somewhere between O & D.
(C) Electric field vanishes two times, first at a point somewhere between A & O and then At O.
(D) Electric field vector changes direction more than once.

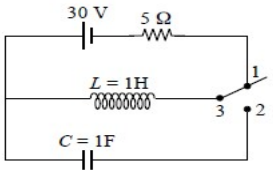
6. If the flux of the electric field through a closed surface is zero.
(A) The electric field must be zero everywhere on the surface
(B) The electric field may be zero everywhere on the surface.
(C) The net charge inside the surface must be zero
(D) The net charge in the vicinity of the surface must be zero.

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

7. A metal sphere of radius a is surrounded by a concentric metal spherical shell of radius $b(b > a)$. The space between the spheres is filled with material whose electrical conductivity σ varies with the electric field strength E according to the relation $\sigma = KE$ where K is constant. A potential difference V is maintained between inner sphere and outer shell. If the current (in ampere) between the inner sphere and outer shell is x, then find $x/4$. [Here $K = \frac{1}{\pi}\times 10^{-2}\ ohm^{-1}\, volt^{-1}, V = 20\, volts, b = ea$ (e = exponential and $\pi = 22/7$]

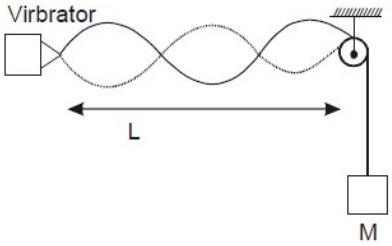
8. In the circuit shown in figure, the switch has been in position '1' for long time. Now the switch is shifted to position 2 at $t=0$. The change on the capacitor is nC at $t=\frac{\pi}{6}$. Find n.



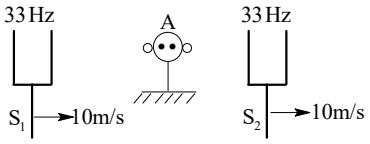
9. A conducting disc of conductivity σ has a radius a and thickness t. If the magnetic field B is applied in a direction perpendicular to the plane of the disc, changes with time at the rat of $\frac{dB}{dt}=\alpha$. Calculate the power dissipated (in mW) in the disc to the induced current.

[Take: $\alpha=2T/s, a=1cm, t=2mm$ and $\sigma=4/\pi\times10^{8}\Omega^{-1}m^{-1}$]

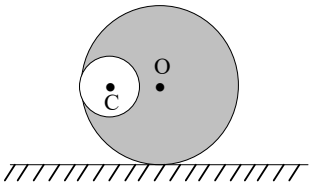
10. In the given arrangement shown in figure, a mass M is suspended with the help of string, which passes over a light pulley. The string is connected to a vibrator having constant frequency. The vibrator gives transverse oscillation to string. When the value of M is either 16 Kg or 25 Kg standing waves are observed, however, no standing waves are observed with any mass between these values. The largest mass for which standing waves could be observed is $\mathrm{n}\times10^{2}\ \mathrm{Kg}$. Find the value of 'n'. (String is very light)



11. Observer A is at rest. Source S_1, is moving towards observer A and source S_2 is moving away from observer A. velocity of sound is $330\ \mathrm{m/sec}$. $\beta=$ beat frequency heard per second by A. Approximate value of β is:



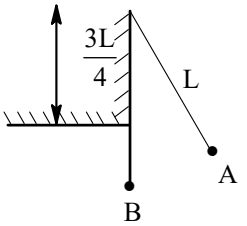
12. A uniform disc of radius $4\mathrm{R}$ has a circular cavity of radius R as shown. The surface mass density of the disc is σ. The disc is held vertically on a rough horizontal surface such that the line joining the geometrical centre 'O' of the disc and centre of the cavity C is horizontal. If the angular acceleration of the disc just when it is released assuming that it starts pure rolling motion and does not topple is $\frac{\mathrm{Ng}}{717\mathrm{R}}$ then find N.



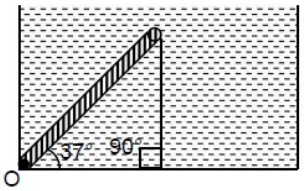
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

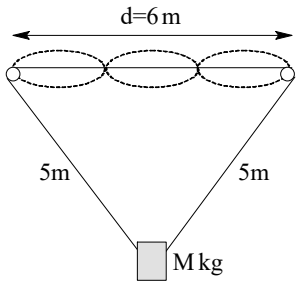
13. A pendulum has period T for small oscillations. An obstacle is placed directly beneath the pivot, so that only the lowest one quarter of the string can follow the pendulum bob when it swings to the left of its resting position as shown in the figure. The pendulum is released from rest at a certain point A. The time taken by it to return to that point is $\frac{\mathrm{nT}}{4}$. Find the minimum value of n ?



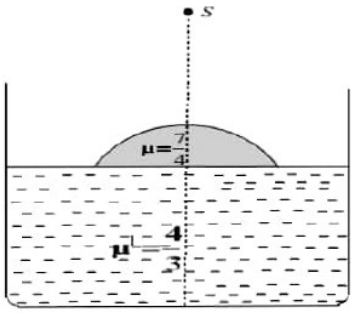
14. A cylindrical rod of uniform cross-section, is attached at O in a water tank. The linear mass density of rod is $\lambda_0 x$, where x is distance of the element of the rod, from end O as shown in figure. If the tension in string is given by $10^4/P$ N then P is (Length of rod 1m, radius of area of cross-section is $\frac{1}{\sqrt{\pi}}$ m, $\rho_{water} = 1000 kg\, m^{-3}, g = 10\, ms^{-2}, \lambda_0 = 10^3$ in S. I. Unit)



15. An object of mass 'M' kg hangs in equilibrium from a string with a total length of L=16 m and linear density of μ=0.01kg/m, as shown in figure. The string is wrapped around two light small frictionless pulleys that are separated by distance d = 6 m. When the string between the pulleys vibrates to form the standing wave pattern with three loops on it, the frequency of vibration is 25 Hz. Find the value of M/2. (Take g = 10 m/s^2)



16. Water (refractive index = 4/3) in a tank is 19 cm deep. Oil of refractive index 7/4 lying on water makes a convex surface of radius of curvature 6cm and acts as a thin lens. An object S is placed 24 cm above the water surface. Its image is formed at $x\,cm$ above the bottom of the tank. Then $x =$ _____



17. The focal length of a thin biconvex lens is $20\,cm$. When an object is moved from a distance of $25\,cm$ to $50\,cm$ in front of it, the magnification of its image changes from m_1 to m_2. The ratio $\frac{m_1}{m_2}$ is_____

18. From the top of tower of height 80 m, a body projected with velocity 50 m/s at an angle of inclination of $37°$. If mass of the body is $m=0.01kg$ and acceleration due to gravity is $10\,m/s^2$, then find power (in watts) supplied by gravitational force on the body seven second after the projection.

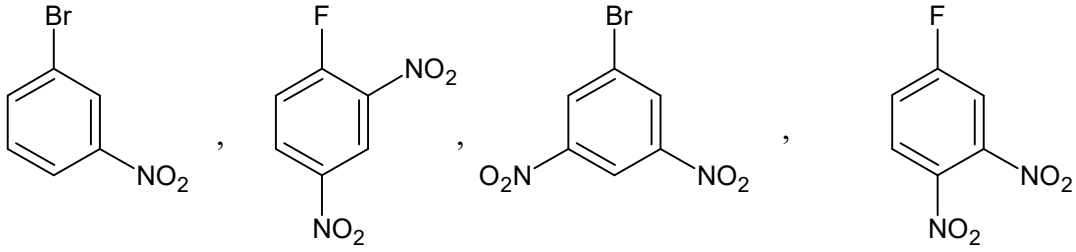
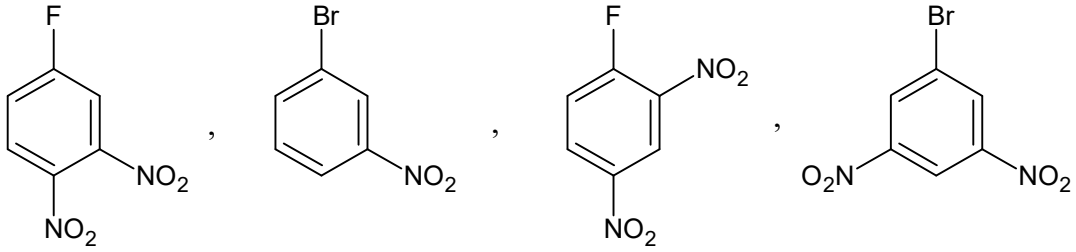
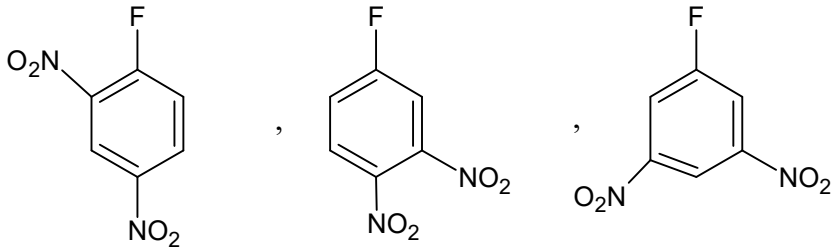
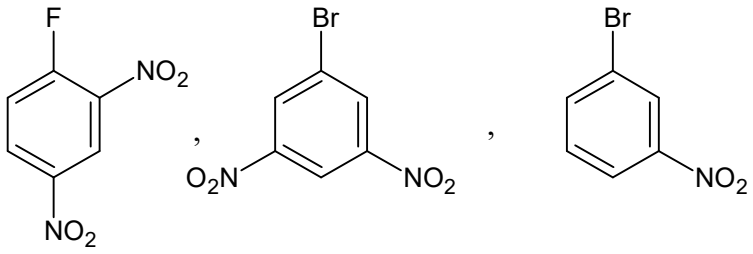
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following statement is/are true?
    (A) Alkali metals are soft metals and softness increases down the group
    (B) Lattice energy of alkali metal salts with same anion decreases with increase in size of cations
    (C) Paramagnetic character of $Na - Liq\ NH_3$ solution decreases in increase in the conc. of NH_3
    (D) LiF has lowest solubility among all alkali metal halides

20. Which of the following correctly ranks the aryl halides in decreasing order of reactivity toward sodium methoxide in methanol?
    (A) 
    (B) 
    (C) 
    (D) 

21. The solubility of alkaline earth metal salts in water decreases down the group, the salts are
(A) Phosphates
(B) Sulphates
(C) Carbonates
(D) Hydroxides

22. In the extraction of Li, a mixture of dry lithium chloride and potassium chloride is fused and electrolysed, the following fact(s) is/are correct regarding this
(A) KCl increases the conductivity
(B) M.P. of LiCl lowers due to presence of KCl
(C) Reduction potential of Li^+ is higher than K^+ in fused state
(D) Reduction potential of Li^+ is lower than K^+ in fused state

23. Which of the following is/are not a characteristic of effusion/diffusion?
(A) The rate of effusion/diffusion is inversely proportional to average speed of the gas particles
(B) Gases mix as a result of the random motion and frequent collision of gas particles
(C) The rate of effusion/diffusion decreases with increasing temperature
(D) ate of effusion/diffusion is directly proportional to molar mass of the gas

24. Which of the following is/are characteristic of a tetrahedral complex?
(A) the $d_{x^2-y^2}$ and d_{z^2} orbitals are the low energy orbitals
(B) most tetrahedral complex are high spin
(C) crystal filed splitting is about twice found in octahedral complexes
(D) the splitting pattern is just opposite of that in octahedral complexes

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

25. Predict whether or not $AgCl$ will be precipitated from a solution which is $0.02M$ in $NaCl$ and $0.1M$ in $[Ag(CN)_2]^-$.
Given $K_{\text{dissociation of the complex ion}} = 4.0\times10^{-17}$ and K_{sp} AgCl$=2.8\times10^{-10}$.
(i) mark 0, if ppt of $AgCl$ form
(ii) mark 1, if no ppt form
(iii) mark 2, if ppt of $NaCN$ form
(iv) mark 3, if ppt of KCN form

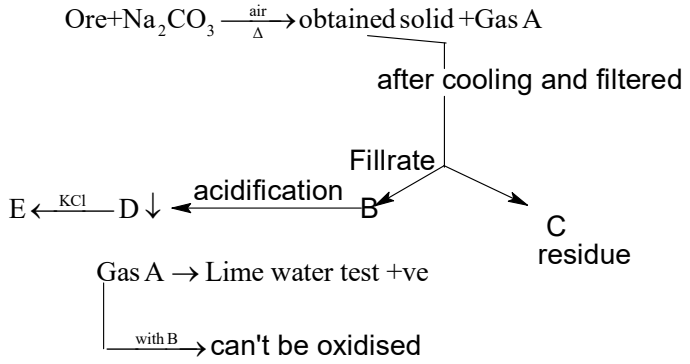
26. A $0.5\,mL$ sample of an aqueous solution containing $0.48\,\mu Ci$ per mL of Na^{22} is injected into the bloodstream of an animal. After allowing sufficient time for complete circulatory mixing a $0.1\,mL$ sample of blood is removed and found to have a net activity of 444 dpm of Na^{22} from this data, the blood volume of animal was found to be $60x\ mL$. Find the value of x.

27. For the two parallel reaction $A \xrightarrow{k_1=2\sec^{-1}} B$ and $A \xrightarrow{k_2=4\sec^{-1}} C$, the activation energy E' for the disappearance of A is given in terms of activation energies E_1 and E_2 for the two path by $E' = \frac{E_1+2E_2}{x}$. The value of x is…….

28. The correct statement(s) is/are $'x'$ then value of $\frac{x}{2.5}$ will be:

(a) In some cases, constitutional isomers are chiral
(b) In every cases, a pair of enantiomers have a mirror-image relationship
(c) Mirror-image molecules are in all cases enantiomers
(d) If a compound has an enantiomer it must be chiral
(e) Every chiral compound has a diastereomer
(f) If a compound has a diastereomer it must be chiral
(g) Every molecule containing one or more asymmetric carbons is chiral
(h) Any molecule containing a stereocentre must be chiral
(i) Any molecule with a stereocentre must show stereoisomerism
(j) Some diastereomers are mirror images to each other
(k) Any chiral compound with a single asymmetric carbon must have a positive optical rotation if the compound has the R configuration.
(l) If a structure has no plane of symmetry it is chiral
(m) All chiral molecules have no plane of symmetry
(n) All asymmetric carbons are stereocentre
(o) Gauche form of n-butane is chiral however anti form is achiral
(p) Molecules with two or more identically substituted stereocenters may exists as meso stereoisomers
(q) A stereocentre can be tetrahedral or trigonal planer

29.



E is an oxidising agent and when a paper dipped in E solution comes in contact with SO_2 changes colour from orange to green.

Now, how many are correct about above flow diagram?

(A) Lime is added with Na_2CO_3, keeps the mass porous so that air becomes accessible to all parts reaction mixture and prevents fusion
(B) 'C' is mainly Fe_2O_3
(C) Filtrate is a yellow solution.
(D) During acidification a redox reaction is taking place.
(E) D contain 2 salts in solution and when fractional crystallisation occurs salt of lower molecular mass comes out first and is filtered off.
(F) Acidication gives two salts both hydrated. The one which has more water of hydration is less soluble
(G) When pH of solution is increased the colour of solution of E changes from orange to yellow
(H) The oxidising anions present in flow diagram can be distinguished by $AgNO_3$ One gives reddish-brown ppt and other gives brick red ppt.

30. $B_3N_3H_6 \xrightarrow{HCl}$ (A)

Number of hydrogen atom is present in compound A after complete addition reactions

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. One mole of an ideal gas is expanded isothermally at 298 K until its volume is tripled. Find the value of ΔS_{sys} (entropy of system) if the expansion is a free expansion (in vacuum). Given (log3= 0.477 R = 8.3 $JK^{-1}mol^{-1}$). Give your answer in JK^{-1} upto two decimals only.

32. Gold metal is extracted from its ore by treating the crushed rock with an aerated cyanide solution. The unbalanced equation for the reaction is

$Au(s) + CN^-(aq) + O_2(g) \rightarrow Au(CN)_2^-(aq)$

Use any of the following data at 27^oC to calculate ΔG° for this reaction in basic solution.

Given K_f for $Au(CN)_2^- = 10^{38}$

$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(\ell)$ $\quad E^\circ = 1.2$ V

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$ $\quad E^\circ = 1.5$ V

$Au^{3+}(aq) + 2e \rightarrow Au^+$ $\quad E^\circ = 1.5$ V

Use equation $\Delta G^\circ = -2RT\log_{10} Q$ and R = 8 $Jmol^{-1}K^{-1}$, F = 96500 C. Give your answer (magnitude only) in kJ.

33. At certain temperature the dissociation constants of formic acid and acetic acid are 1.8×10^{-4} and 1.8×10^{-5} respectively. The concentration of acetic acid solution(molarity) in which the hydrogen ion has the same concentration as 0.001 M formic acid solution is equal to

34. If 250 ml O_2 over water at 30°C and a total pressure of 740 torr is mixed with 300 ml of N_2 over water at 25^oC at total pressure of 780 torr. What will be total pressure if the mixture is in a 500 ml vessel over water at $35°C$. (Given aq. tension of water at 25^oC, 30^oC and 35^oC are 23.8, 31.8 and 42.2 torr.)

35. The solubility product of $Co_2[Fe(CN)_6]$ in water is $K\times10^{-16}$. The value of K is

Given $\lambda^\circ_m Co^{+2} = 80\Omega^{-1}$ cm^{-1} mol^{-1}

$\lambda^\circ_m[Fe(CN)_6]^{-4} = 40\Omega^{-1}$ cm^{-1} mol^{-1}

Specific conductivity of saturated solution $= 2\times10^{-6}\Omega^{-1}c\ m^{-1}$

Specific conductivity of water used = $= 4\times10^{-7}\Omega^{-1}c\ m^{-1}$

36. In a eudiometer 14 ml of CH_4 and 38.5 ml of O_2 was introduced. After explosion and cooling to initial temperature (23^oC) the volume of eudiometer was found to be 25 ml. Calculate the vapour pressure (in mm of Hg) of aqueous solution containing a non-volatile solute (A) at 23^oC. Mole fraction of (A) is 0.1. The pressure of eudiometer is 1 atm and is constant.

Mathematics

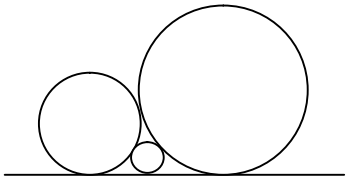
PART – III

SECTION – A
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $\sqrt{\sin^{-1}|\sin x|}+\sqrt{\tan^{-1}|\tan x|}=2\sqrt{a}$ has atleast one solution; then 'a' belongs to
(A) $\{0\}$
(B) $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$
(C) $\left[0,\ \frac{\pi}{2}\right)$
(D) $\left[\frac{-\pi}{2},\ 0\right)$

38. Two circles with radii 1 and 2 are placed so that they are tangent to each other and also touch a straight line (as shown in fig). A third circle is nestled between them that it is tangent to the first two circles and the lines. Then the radius of third circle is



(A) $6-4\sqrt{2}$
(B) $6+4\sqrt{2}$
(C) $\frac{2}{3-2\sqrt{2}}$
(D) $\frac{2}{3+2\sqrt{2}}$

39. f, g, h : R $\to$ R, f(x), g(x), h(x) are all continuous, differentiable functions which satisfy the relation f(x + y) = g(x) + h(y) for all x, y $\in$ R
(A) f′(0) = f′(1)
(B) g′(x) = h′(x) + 1
(C) h(x) $\neq$ g(x) $\forall$ x $\in$ R – {0} if h(0) $\neq$ g(0)
(D) f(0) = g(0) + h(0)

40. Consider the equation $\sqrt{1+\frac{3\cos 2x}{\sqrt{\sin^4 x+4\cos^2 x}+\sqrt{\cos^4 x+4\sin^2 x}}}$ = $\sqrt{2}\sin^{-1}(\sin 2x)$ x $\in$ [– π, π)
(A) Number of roots of this equation is 2
(B) Number of roots of this equation is 4
(C) One root lies in the interval $\left(\frac{\pi}{8},\frac{\pi}{6}\right)$
(D) If x $\in$ [0, nπ] n $\in$ N, number of roots will never be odd.

41. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, then

(A) $A^3 - A^2 = A - I$

(B) Det $(A^{2010} - I) = 0$

(C) $A^{50} = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$

(D) $A^{50} = \begin{bmatrix} 1 & 1 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$

42. Let f(x) and g(x) are two non periodic functions then which of the following may be periodic functions

(A) f(x).g(x)

(B) f(x) + g(x)

(C) f(x) – g(x)

(D) f(g(x))

SECTION – B
(Single Digit Integer Type)

This section contains **06** questions. The answer to each question is a **Single Digit integer** ranging from **0** to **9**, both inclusive.

43. Value of $1.(2 - w)(2 - w^2) + 2.(3 - w)(3 - w^2) + 3.(4 - w)(4 - w^2) + 4.(5 - w)(5 - w^2) = 44k$, k = _____ (w is the cube root of unity).

44. If $x \in (1°, 30°)$, $(1 + \tan(x)°)(1+ \tan(x + 1)°)(1 + \tan(x + 2)°) \ldots (1 + \tan(x + 44)°) = 2^{23}$. Value of x is _____ degrees.

45. X wants to put 5 different rings into three different pockets is that no pocket is empty. Then the probability of such event occurring is $\frac{1}{P}$. [P] = _____ ([.] is the greatest integer function).

46. $\left|\left[\int_2^3 \frac{dx}{\sqrt{1+x^3}}\right]\right|$ = _____, (where [.] denotes the greatest integer function, |.| denotes absolute value function).

47. The total number of common tangents drawn to all possible pair of the circles $x^2 + y^2 + 4y - 1 = 0$, $x^2 + y^2 + 6x + y + 8 = 0$ and $x^2 + y^2 - 4x - 4y - 37 = 0$ are

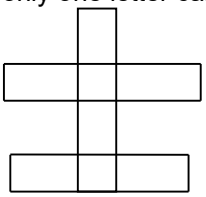
48. If the coordinates (x, y, z) of the point S which is equidistant from the points O(0, 0, 0), A(n^5, 0, 0,) B(0, n^4, 0), C(0, 0, n) obey the relation $2(x + y + z) + 1 = 0$. If $n \in z$, then |n| = _____ (|.| is the modulus function)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. A plane P contains the line $L_1 : \frac{y}{b}+\frac{z}{c}=1, x=0$ and is parallel to the line $L_2 : \frac{y}{a}-\frac{z}{c}=1, y=0$. If the shortest distance between L_1 and L_2 is $\frac{1}{4}$ then the value of $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ is equal t_____

50. Six different letters are placed in boxes of the given figure such that no row is empty. (In each box only one letter can be placed). The member of such arrangements is__



51. If $f(x)=x+1$, $g^{-1}(x)=x^3+x+1$ and $g(f(x))=h(x)$, then $\lim\limits_{x\to 1}\frac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to____

52. If $P(A\cup B)=\frac{3}{4}$; $P(A\cap B)=\frac{1}{4}$ $P(\overline{A})=\frac{2}{3}$ then $P(\overline{A}\cap B)$ is_____

53. Number of rational terms in expansion of $\left(\sqrt{2}+\sqrt{3}+\sqrt[3]{5}\right)^{20}$ is a two digit number ab, then a + b =

54. Let A = {1, 2, 3, 4, 5} and B = {0, 1, 2, 3, 4, 5}, how many one–one function f: A $\to$ B can be define so that f(i) $\neq$ i $\forall$ i = 2, 3, 4, 5 and f(1) $\neq$ 0 or 1